

NELSON MANDELA UNIVERSITY

DEPARTMENT OF BUSINESS MANAGEMENT EBMV301 FINANCIAL MANAGEMENT SEMESTER TEST 2 (TM2) APRIL 2026

MEMO

OPEN QUESTIONS

[60]

QUESTION 1

[13]

1.1

(10)

Step 1: Calculate dividends during variable (first) growth period

$$D_0 = R3.20 \text{ (given)}$$

$$D_1 = D_0(1 + g) = 3.20(1 + 0.12) = 3.58 \quad (1)$$

$$D_2 = D_1(1 + g) = 3.58(1 + 0.12) = 4.01 \quad (1)$$

Step 2: Sum the PV of these dividends during variable (first) growth period

$$D_1 = FV = -R3.58; \quad n = 1; i = 13 \quad PV = R \ 3.17 \quad (1)$$

$$D_2 = FV = -R4.01; \quad n = 2; i = 13; \quad PV = \underline{R \ 3.14} \quad (1)$$

$$\underline{\underline{R \ 6.31}}$$

Step 3: Value after (at end of) variable period (thus second period now)

$$\begin{aligned} D_3 &= D_2(1 + g) \\ &= 4.01(1 + 0.06) \\ &= R4.25 \end{aligned} \quad (1)$$

$$P_2 = D_3 / (k - g) \quad (1)$$

$$\begin{aligned} &= 4.25 / (0.13 - 0.06) \\ &= R60.71 \end{aligned} \quad (1)$$

Step 4: PV after (at end of) variable period (thus second period now)

$$FV = -R60.71 \quad n = 2 \quad i = 13 \quad PV = 47.54 \quad (1)$$

Step 5: Sum the PVs in steps 2 and 4

$$6.31 + 47.54 = R53.85 \quad (1)$$

Therefore, 557 (since $30\ 000 / 53.85 = 557.10 = 557$) shares can be purchased.

(1)

1.2 (1)
PMT = -80; FV = -1 000; PV = 1 150.75; I = 6.5
Compute n = 16.82 years

1.3 (2)
Premium (1)
The bond is trading at a price higher than its face value (R1 150.75 > R1 000) OR
The coupon rate (8%) is higher than the yield to maturity (6.5%) (1)
(making the bond more attractive to investors)

QUESTION 2 [13]

2.1 (4)
Should infer Inventory Conversion Period (ICP) = 70 (1)

Cash conversion Cycle (CCC) = AAI + ACP - APP (1)
= (70 + 45) - 30
= 85 days (1)

On average, there is 85 days between cash disbursements (outflows) and cash inflows (1)

2.2 (1)
A cash gap/shortage can be financed by means of short term (ST) borrowings or using liquidity reserves.

2.3 (2)
Lengthen (1)
Negotiate (better/extended) credit terms with suppliers/creditors (1)

2.4 (2)
Cost of forfeiting cash discount = $\left[\frac{3}{100-3} \right] \times \left[\frac{360}{30-10} \right]$ (1)
= 55.67% p.a. (1)

2.5

Solution showing answers to questions 2.5.1 and 2.5.2:

**SUNSTREAM COMPANY
CASH BUDGET FOR JUNE 2026**

	R
	June
Cash receipts	810 000
Other income	12 000
Total receipts	822 000
Payment for purchases	650 000
Lease	20 000
Salaries and wages	70 000
Dividends	-
Loan repayments and interest	15 000
Tax	5 000
Total disbursements	760 000
NET CASH FLOW	62 000
Cash at the beginning of the period or minimum	100 000
Cash at the end of the period before borrowing	162 000

2.5.1 D (2)

Payment for May purchases is made in June

2.5.2 C (2)

See cash budget above

QUESTION 3**[9]**

3.1

(2)

Credit cost $= R4\ 500 \times 0.12 \times 60 / 360$
 $= R90.00$ (1)

Cost price	R4 500.00	
Credit cost	<u>R 90.00</u>	
Total cost	<u>R4 590.00</u>	(1)

3.2

(2)

Sales price	R6 200.00	
Less cost	<u>R4 590.00</u>	
Profit	<u>R1 610.00</u>	(2)

3.3

(2)

Credit cost $= R4\ 500 \times 0.12 \times 15 / 360$
 $= R22.50$ (1)

Cost price	R4 500.00	
Credit cost	<u>R 22.50</u>	
Total cost	<u>R4 522.50</u>	(1)

3.4

(3)

Sales price $= \text{Sales price minus discount}$
 $= R6\ 200 - (R6\ 200 \times 0.03)$
 $= R6\ 014.00$ (1)

Sales price	R6 014.00	
Less cost	<u>R4 522.50</u>	
Profit	<u>R1 491.50</u>	(1)

No, Apex should not implement the new credit terms as the profit will decrease. (1)

QUESTION 4**[10]**

4.1

(3)

$$EOQ = [(2 \times F \times S) / (C \times P)]^{0.5} \quad (1)$$

$$= [(2 \times R120 \times 60\,000) / 5]^{0.5}$$

$$= 1\,697.06 \text{ units}$$

$$= 1\,698 \text{ units} \quad (1)$$

(cannot order a part of a unit, cannot round down, as the company will run out of stock at some point in time)

The EOQ is 1 698 units – number of units ordered each and every time (1)

4.2

(3)

$$ROP = \text{Lead time in days} \times \text{Daily sales} \quad (1)$$

$$= 8 \text{ days} \times 165 \text{ units}$$

$$= 1\,320 \text{ units} \quad (1)$$

When the stock level reaches 1 320 units, a reorder (of 1 698 units) must be placed. (1)

4.3

(2)

$$TCC = C \times P \times A \quad (1)$$

$$= R5 \times 845 \text{ units}$$

$$= R4\,225 \quad (1)$$

4.4

(2)

Any two for 1 mark each (each discussed in a full sentence, not to be listed; if only listed: ½ mark each):

Ordering costs

Carrying costs

Shortage costs

QUESTION 5**[15]**

5.1

(13)

	YEAR 0	YEAR 1	YEAR 2	YEAR 3	YEAR 4
Initial investment outlay	-3 800 000 (2)				
Operating cash flow		1 507 000 (½)	1 507 000 (½)	1 507 000 (½)	1 507 000 (½)
Terminal cash flow					648 200 (2)
Total cash flow per period	-3 800 000 (½)	1 507 000 (½)	1 507 000 (½)	1 507 000 (½)	2 155 200 (1)

NPV CF calculator inputs shown and/or YTM shown as 16% (1)
NPV = R774 853.31 (1)
Accept project (1)
NPV is positive (1)

Workings:

Initial investment outlay = Purchase price + change in NWC (1 mark if figure incorrect above)
= -R3 400 000 – R400 000
= -R3 800 000

Operating cash flow

OCF = EBIT + Depreciation – Taxes (worth 3½ above, so if figure incorrect: 1 mark for equation)
= 900 000 + (3 400 000 / 4) – 243 000
= R1 507 000

Terminal cash flow

After tax value of proceeds = Selling value of equipment x (1 – tax) (1 mark if figure incorrect above)
= 0.10 x R3 400 000 x (1 – 0.27)
= R248 200 (1 mark if figure incorrect above)

Terminal cash flow = After tax value of proceeds + reverting back of change in NWC
= R248 200 + R400 000
= R648 200 (if figure incorrect: ½ mark for equation)

5.2 C (2)

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